

EasySave v3.0

Release Note

ProSoft — v3.0 — 2026

1. Overview

EasySave v3.0 is a major release introducing parallel backup execution, real-time job control (Pause/Play/Stop), priority file management, large file transfer limiting, CryptoSoft mono-instance enforcement, and centralized log management via Docker.

2. What's New

Parallel backup

- All backup jobs now run simultaneously via Task.WhenAll()
- Each job runs in its own Task — the UI remains responsive
- Sequential mode is no longer available

Priority file management

- User defines priority extensions in Settings (e.g. .pdf, .docx)
- Priority files are copied before all other files
- Managed by the new static PriorityQueue class with thread-safe lock

Large file transfer limit (SemaphoreSlim)

- User configures a max parallel file size threshold in Ko (default: 15,360 Ko)
- Only one file above the threshold can be transferred at a time
- Files below the threshold transfer freely in parallel

Pause / Play / Stop per job

- Each job has its own BackupJobController
- Pause: waits for the current file to complete, then blocks
- Play: resumes from where it stopped
- Stop: cancels immediately via CancellationToken

Business software — auto-pause (v2.0 behaviour updated)

- v2.0: backup stopped when business software detected
- v3.0: all jobs pause automatically, resume when software closes
- Background monitoring thread checks every 1000ms

CryptoSoft mono-instance (Mutex)

- CryptoSoft.exe now uses a global Mutex: Global\CryptoSoftMutex
- Only one instance can run at a time across all parallel jobs
- CryptoSoftService waits for the Mutex before launching encryption

Docker — centralized log server

- New ASP.NET Core minimal API server containerized with Docker
- EasySave sends each log entry via HTTP POST to the server
- 3 log destination modes: Local / Remote / Both
- One daily log file regardless of the number of machines

- New RemoteLogFormatter sends entries via HttpClient

3. New Files

File	Action	Description
BackupJobController.cs	NEW	Pause/Play/Stop via CancellationToken + ManualResetEventSlim
BackupSemaphore.cs	NEW	SemaphoreSlim(1,1) to limit simultaneous large file transfers
PriorityQueue.cs	NEW	Static class — manages priority file extensions with thread-safe lock
RemoteLogFormatter.cs	NEW	Sends log entries to Docker server via HttpClient HTTP POST
LogServer/Program.cs	NEW	ASP.NET Core minimal API — POST /log, GET /health, GET /logs
Dockerfile	NEW	Docker image for the centralized log server

4. Modified Files

File	Action	Change
BackupJob.cs	MODIFY	ExecuteAsync — parallel Task, priority sort, pause/stop checks, semaphore
BackupManager.cs	MODIFY	RunAll() replaced with Task.WhenAll() parallel execution
BackupJobViewModel.cs	MODIFY	Added PauseCommand, PlayCommand, StopCommand
MainWindow.xaml	MODIFY	Added Pause/Play/Stop buttons and progress bar per job
MainViewModel.cs	MODIFY	Added StartMonitoringThread() for business software detection
BusinessSoftwareService.cs	MODIFY	IsBlocked static flag — auto-pause instead of stop
CryptoSoftService.cs	MODIFY	Global Mutex Global\CryptoSoftMutex for mono-instance
CryptoSoft/Program.cs	MODIFY	Mutex at startup — rejects second instance
Settings.cs	MODIFY	Added PriorityExtensions, MaxParallelFileSizeKo, LogDestination, DockerServerUrl
Logger.cs	MODIFY	Added SetRemoteFormatter(), SetDestination() — Local/Remote/Both modes
SettingsWindow.xaml	MODIFY	Added Docker URL, log destination, priority extensions fields

5. Version Comparison

Feature	v2.0 / v3.0
Backup execution	v2.0: sequential / v3.0: parallel (Task.WhenAll)
Business software	v2.0: stops backup / v3.0: auto-pause + auto-resume
CryptoSoft	v2.0: no restriction / v3.0: mono-instance (Mutex)
Pause/Play/Stop	v2.0: no / v3.0: yes (per job)
Priority files	v2.0: no / v3.0: yes (PriorityQueue)
Large file limit	v2.0: no / v3.0: yes (SemaphoreSlim)
Log centralization	v2.0: local only / v3.0: Local/Remote/Both (Docker)

